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MAGNETIC POLARITY STRATIGRAPHY OF THE EARLY TERTIARY BLACK PEAKS FORMATION, BIG BEND NATIONAL PARK, TEXAS[1]

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ABSTRACT

The fossiliferous Black Peaks Formation consists of fluviatile mudstones, silts, and sands. The faunal succession of this deposit indicates an age ranging from middle Paleocene (Torrejonian) to early Eocene (Clarkforkian). A minimum of three separately oriented paleomagnetic samples were collected from each of 53 sites in the type and one reference section. Depending upon pilot studies and lithology, in the laboratory the paleomagnetic specimens were treated with either AF or thermal demagnetization (at, respectively, a minimum regime of 25 mT or 200° and 400°C), which appeared to isolate the characteristic component of magnetization. The magnetic polarity zonation is predominantly reversed punctuated by several normal intervals. Based on the faunal constraints and magnetostratigraphic pattern, the Black Peaks is correlated to the reversed zone below chron 26, to the base of chron 24, or an absolute interval of time ranging from about 62 to 53 Ma. Previous hypotheses that the Javelina-Black Peaks and Black Peaks-Hannold Hill lithological boundaries approximate, respectively, the Cretaceous-Paleocene and Paleocene-Eocene boundaries, need to be revised in light of the present study. Rates of sediment accumulation represented in the Black Peaks Formation appear to have been relatively constant with a mean value of about 1.8 cm/1000 yrs. The transitions between Torrejonian-Tiffanian and Tiffanian-Clarkforkian faunas occurs in respectively, the reversed zones below chron 26 and chron 24, consistent with the situation in the Clark's Fork Basin of Wyoming.

INTRODUCTION

The Black Peaks Formation of Big Bend National Park, Texas, consists of continental sedimentary deposits containing several significant Paleocene and early Eocene faunas. The geologic history of this formation is important because it is the only Paleocene sedimentary rock unit found in the Trans-Pecos region, and it also contains the southernmost occurrence of large Paleocene mammalian faunas so far discovered in North America (Schiebout 1974, 1980). The purpose of this report is to present the chronology of the Black Peaks Formation by the establishment of a magnetic polarity stratigraphy for this unit. On a broader scale, comparisons will then be made with other roughly contemporaneous faunas which will provide a better understanding of early Tertiary biostratigraphy and paleobiogeography.

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In western North America, continental Paleocene faunas are relatively abundant (Schiebout 1980). Several recent studies have presented the magnetic polarity stratigraphy of some of these classic localities. Butler et al. (1977) and Lindsay et al. (1978) present the magneto- and biostratigraphy of the richly fossiliferous Paleocene Nacimiento Formation of the San Juan Basin, New Mexico. They state that Puercan (early Paleocene) and Torrejonian (middle Paleocene) faunas correlate to, respectively, chrons 28 and 26 of the Magnetic Polarity Time Scale (MPTS). Tomida and Butler (1980) present the magnetostratigraphy of the North Horn Formation of central Utah. This unit, which contains the problematical early Paleocene Dragon Local Fauna (L.F.), correlates to chron 27 and the reversed interval between chrons 27 and 26 (also see Tomida 1981). Butler et al. (1981) present the magnetostratigraphy of the classic Tiffanian (late Paleocene) to early Eocene Clarkforkian (*sensu* Rose 1981) sequence in the Big Horn Basin of Wyoming, which correlates to the interval preceding chron 26 to the interval preceding chron 24 of the MPTS.

The abundance of relatively fine-grained sediments and long, continuous exposures

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with associated vertebrate-bearing horizons makes the Black Peaks Formation amenable to an interdisciplinary study similar to those already reported for other Paleocene localities. With the rich suite of late Paleocene to early Eocene mammals reported from the Black Peaks Formation (Schiebout 1974), the present study should clearly fall within an interval on the MPTS similar to those already presented above for the Paleocene.

GEOGRAPHIC AND GEOLOGICAL SETTING

The area that is now Big Bend National Park of Brewster County, Texas (fig. 1), was first described by Udden (1907) as a sunken block, which created a basin of Tertiary sedimentary and volcanic rocks. The boundaries of this 65 km wide basin are the Mesa de Angulia (a northwest trending tilted fault block), the Terlingua Fault Zone and Terlingua Uplift to the southwest, and Sierra del Carmen-Santiago ranges (a series of fold and fault block ridges) to the east. Although there are numerous geological publications on this region, Maxwell et al. (1967) provide one of the best summaries relevant to the present report.

The Big Bend area is comprised of igneous, metamorphic, and sedimentary outcrops that range in age from Paleozoic to Recent. The total thickness of sedimentary and volcanic rocks that are deposited in this area is 4,200 m. Of relevance to the present study, the late Cretaceous and early Tertiary sedimentary

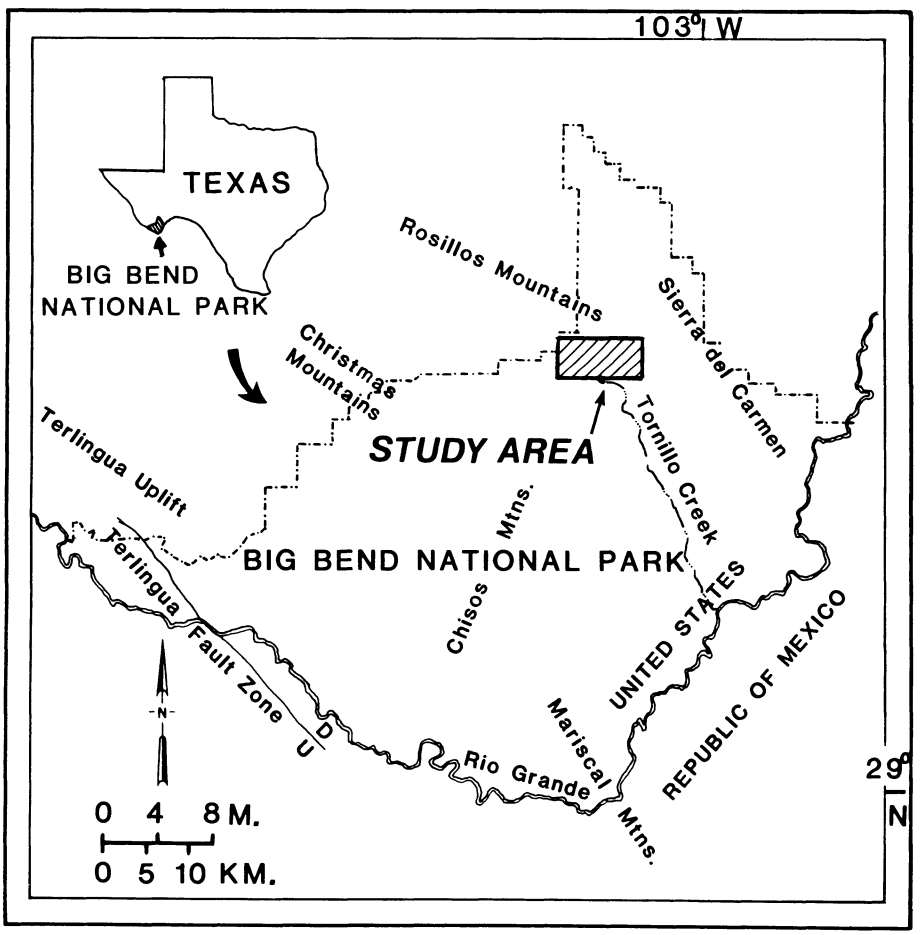


FIG. 1.—Location of Big Bend National Park, Trans-Pecos, Texas. Study area of Black Peaks Formation on eastern and western Tornillo Flat is indicated by shaded region in the northern part of the park. For a detailed geological map of this region, see Maxwell et al. (1967) and Schiebout (1974).

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rocks record a regressive sequence characterized by the following units: the Boquillas Formation (carbonate shelf deposits); the Aguja Formation (deltaic and near shore marine deposits); and fluvial deposits of the Tornillo Group, which includes the Javelina, Black Peaks, and Hannold Hill formations. The Hannold Hill Formation shows the first sedimentological evidence of local tectonism (Hartnell 1980), and the upper Canoe Formation shows the first evidence of local volcanism (Rigsby 1982). The Quaternary deposits principally consist of undifferentiated alluvium.

The Black Peaks Formation (Maxwell et al. 1967) takes its name from three small basaltic peaks exposed on eastern Tornillo Flat. Sediments referred to this unit investigated during the present study crop out on both eastern and western Tornillo Flat (fig. 1). The Black Peaks Formation consists of a maximum of about 180 m of gray, maroon, and black mudstones and gray buff, or yellow sandstones that were deposited in a fluviatile paleoenvironment. Prominent color banding of mudstones is a result of changes in soil-forming processes caused by major shifts in the water table level. Red layers rich in soil-formed microcrystalline calcium carbonate nodules developed in relatively dry periods, and black layers, colored by organic matter, formed when the water table was at, or above, the floodplain surface in wet periods. Climate and base level changes that produce the banding certainly affected rates of deposition. However, the rates of sedimentation that are calculated below are figured at a large enough scale not to have been affected by these shorter-period perturbations (Schiebout 1970, 1974, 1981; Schiebout and Ting 1981). The sediments are relatively undeformed and are exposed in monoclinal cuestas with shallow dips (less than 5 to 8°) in a characteristic badlands topography. The Black Peaks Formation is underlain by the predominantly Cretaceous Javelina Formation and is overlain by, depending on the locality, (1) the Hannold Hill or Canoe formations, both of which are of Eocene age, or (2) Pleistocene alluvium. Early Tertiary fossil vertebrates were first reported from the Tornillo Flat area by Wilson et al. (1952). Subsequently, numerous other localities have been collected in this continental sequence, particularly in the Black Peaks Formation (Schiebout 1974).

FIELD AND LABORATORY PROCEDURES

At least three separately oriented hand samples were collected from 53 magnetic sites (horizons). Of these, 43 were collected from the type section of the Black Peaks Formation on western Tornillo Flat. The remaining 10 sites were collected along the lower part of reference section 1 of this study. The average vertical separation between paleomagnetic sites is about 5 m. These sites are keyed to the stratigraphic sections that were measured during this study and also those published in Schiebout (1974). Because fine-grained deposits characteristically yield the most accurate detrital remanent magnetization (DRM) results (e.g., Verosub 1977; Cain et al. 1979), mudstones and siltstones were selected for sampling. Wherever possible, sampling of maroon beds was avoided because of potential problems resulting from chemical remanent magnetization (CRM).

In the laboratory, the field samples were cut to standard dimensions (1.64 cc) using a belt sander. Magnetic measurements, demagnetizations, and data reductions and analyses were done at the University of Florida Paleomagnetic Laboratory. This facility includes a Superconducting Technology 2-axis cryogenic magnetometer (Goree and Fuller 1976). Samples were demagnetized by either a Schonstedt GSD-1 AF Geophysical Specimen Demagnetizer or a Schonstedt TSD Thermal Specimen Demagnetizer. Calculations of magnetic vectors and Fisher statistics were performed on a Pet 2001 Series Commodore BASIC Personal Computer and line printer.

PALEOMAGNETIC MEASUREMENTS AND DATA ANALYSIS

In order to better understand the magnetic characteristics of the Black Peaks sediments, pilot demagnetization studies were performed on selected paleomagnetic specimens. Because there is a range of sedimentary lithologies and paleoenvironments represented (see above), experiments were done with both AF and thermal demagnetization. Specimens of gray and buff mudstones and silty mudstones were subjected to successive AF demagnetizations of 5, 10, 15, 20,

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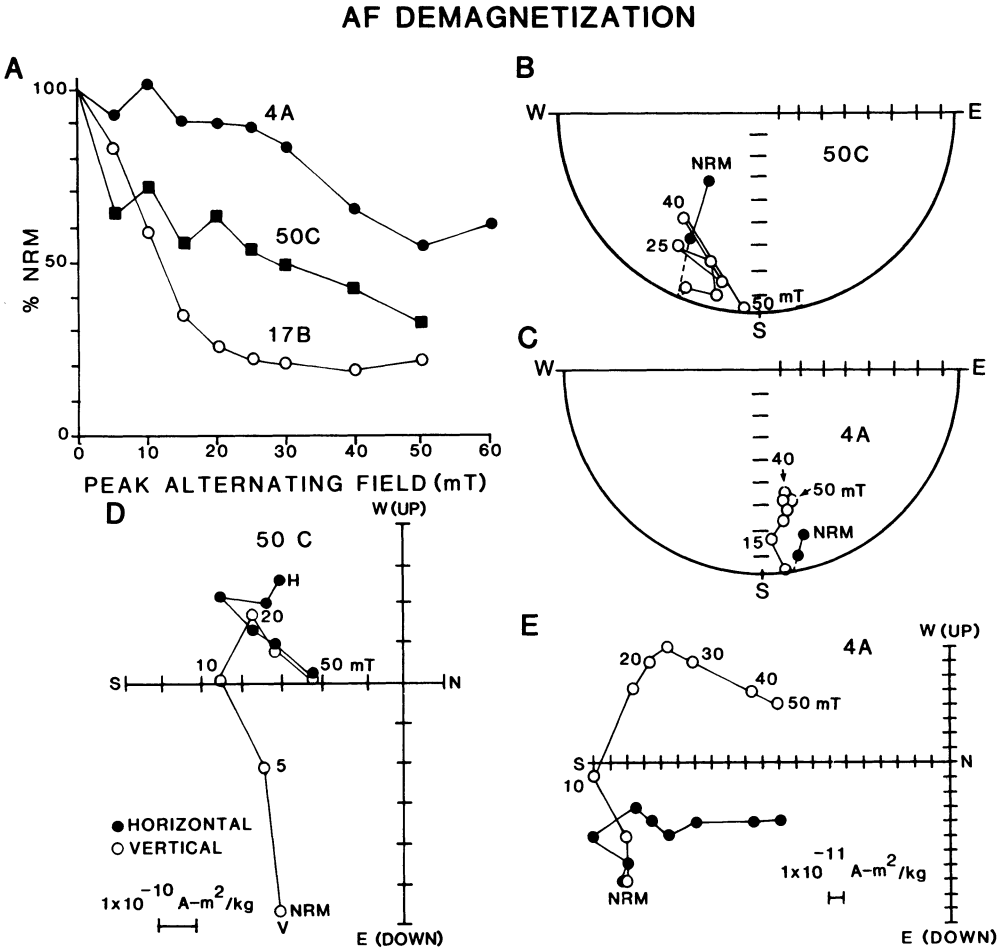


FIG. 2.—AF demagnetization characteristics for representative specimens from the Black Peaks Formation, 4A, 17, and 50C. *A*. Change in intensity (expressed as % NRM) at successively higher alternating fields. *B, C*. Stereographic projections. Closed and open circles represent, respectively, positive and negative inclinations. *D, E*. Vector demagnetization diagrams. Closed and open circles represent, respectively, horizontal and vertical components.

25, 30, 40, 50, and in a few cases, 60 mT. Maroon mudstones were subjected to thermal demagnetizations at increments of 100° or less up to, depending on the particular specimen, maximum temperatures between 600° and 700°C.

NRM intensities for all sites have a mean value of about 2×10^{-9} A-m^2/kg. There is no systematic difference in these intensities that can be correlated to lithological variations.

The results of the pilot AF demagnetizations are represented in figure 2. For the NRM, 50 mT, and 100 mT, a low coercivity component is represented by the directions with positive inclinations in the stereographic projections (figs. 2*B* and 2*C*), as well as the horizontal and vertical components (figs. 2*D* and 2*E*). Once this normal overprint, which is presumably viscous in origin, is removed, there are apparent stable directions in both the stereographic projections for 20–40 mT (figs. 2*B* and 2*C*) and vector demagnetization diagrams (figs. 2*D* and 2*E*). In most cases these samples suggest isolation of a characteristic magnetic component of reversed polarity as a result of demagnetization in alternating fields of at least 20 mT. (The shallow positive inclinations shown at 50 mT for specimen 50C probably resulted from the acquisition of a spurious magnetization in the laboratory during demagnetization in a relatively high field.)

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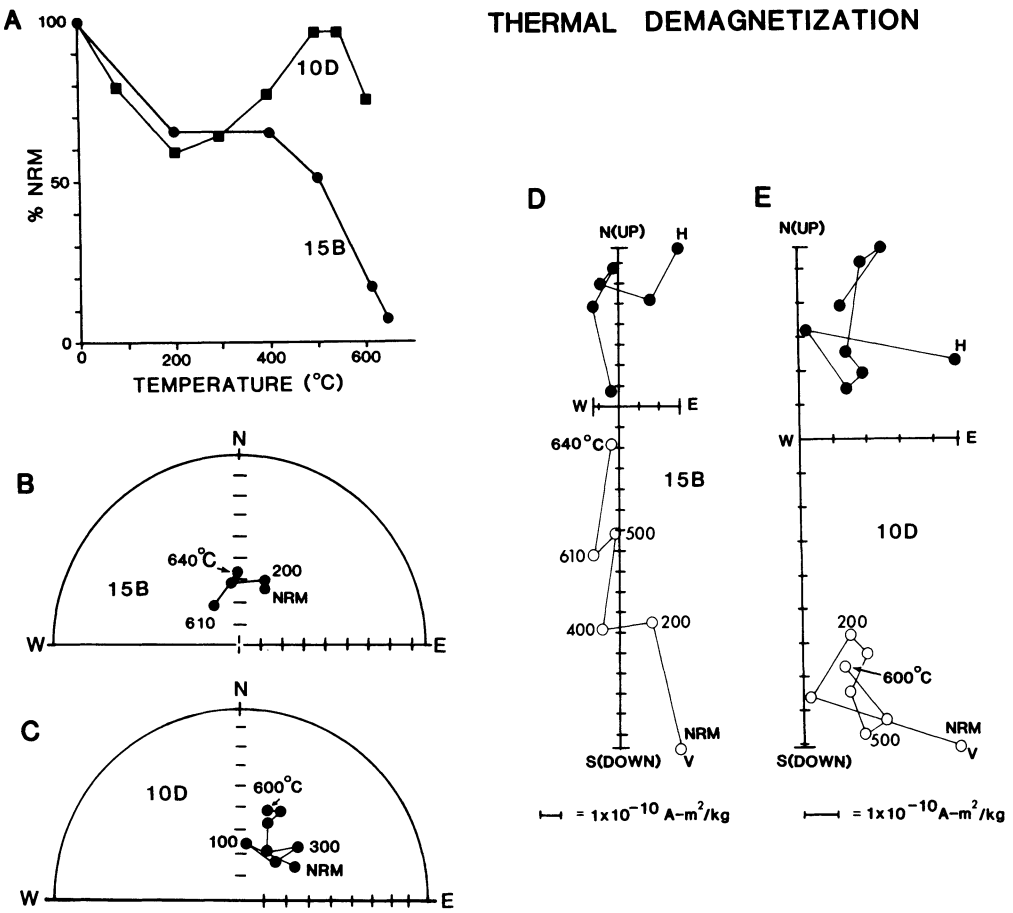


FIG. 3.—Thermal demagnetization characteristics for representative specimens from the Black Peaks Formation, 10D and 15B. *A*. Change in intensity (expressed as % NRM) at successively higher temperatures. *B, C*. Stereographic projections. *D, E*. Vector demagnetization diagrams. Conventions are the same as for figure 2.

As a result of these pilot AF studies, all specimens of the gray or tan (i.e., not obviously oxidized) sediments, which represented the majority of sites (32), were routinely demagnetized at 25 mT. Many of these same specimens also were demagnetized at higher fields (40, 50, 60 mT) in order to determine if stable magnetic vectors had indeed been isolated. As discussed below (under statistical analysis), specimens treated at higher fields either exhibited improved within-site precision or did not significantly change relative to data obtained at 25 mT.

In addition, specimens from seven sites were treated with thermal demagnetization at relatively low temperatures (i.e., 200°C and sometimes 400°C) after having been treated with AF demagnetization. In some cases this combination of regimes improved within-site data sets, whereas in other cases, there was no significant change. One of the most obvious differences seen in the improved data sets treated in this way is steepening of inclinations to values closer to those "expected" for an antipodal distribution (for normal and reversed populations). These changes could be explained by the removal of a portion of the NRM resulting from the presence of sedimentary goethite or a thermally discrete, low blocking temperature component.

The maroon, apparently oxidized, mudstones were subjected to thermal demagnetization. Figure 3 shows the representative characteristics of this sediment type as a result of this regime. In both 10D and 15B (fig. 3), the stereographic projection indicates a vector of normal polarity. Although specimens 10D and 15B do not show a simple

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linear decay toward the origin, the vector demagnetization diagrams do show that the horizontal and vertical components remain in, respectively, the northern and down quadrants, indicating normal polarity. These demagnetization data show that there is a contribution to the magnetization above the Curie Point of magnetite, indicating the presence of hematite as a carrier of the remanence. It is not clear whether hematite or magnetite (or both) is the principal magnetic mineral at lower temperatures. However it is important to note that the direction of magnetization of these specimens is the same both above and below the known Curie Point of magnetite (i.e., for stable end-members of the titanomagnetite solid-solution series). In summary, the NRM of the maroon mudstones results from at least CRM and possible DRM. Samples from 10 sites were subjected to thermal demagnetization at 200°C, 400°C, and in some cases 610°C.

A few specimens of maroon mudstones were also subjected to AF demagnetization. However, the results of these experiments suggested, as might be expected for oxidized sediments, that AF demagnetization was not effective in isolating a characteristic component of the NRM.

As will be discussed in the next section, the combined AF and thermal demagnetization regimes apparently were successful in isolating the characteristic magnetization and consequent interpretation of site polarities. Generally, the AF demagnetized mudstones seemed to yield better statistical data than either the AF demagnetized coarser sediments or the thermally demagnetized mudstones.

The paleomagnetic data for all sites were analyzed using Fisher's statistics (Fisher 1953). For all sites, statistical parameters were calculated for the NRM and their optimum demagnetization. As discussed above, some specimens were demagnetized two or more times at successively higher fields or temperatures. In some cases the additional demagnetization improved within-site clustering, in other cases not. The data set with the greatest precision was chosen for the Fisher calculations. Only sites whose mean directions are significant at the 95% confidence level (i.e., $R \geq 2.62$ for $N = 3$; $R \geq 3.10$ for $N = 4$; see Irving 1964, p. 68) were used to illustrate these vector positions on a stereographic projection and to calculate an overall mean declination and inclination for sites of normal and reversed polarity. A few sites that are statistically significant but of dubious polarity (i.e., positive inclinations and declinations near 180° or those with intermediate directions) are not included in figure 4 and the following discussion.

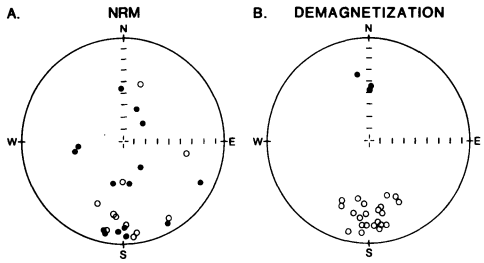


FIG. 4.—Stereographic projections of statistically significant site means for the Black Peaks Formation (including the type and reference sections). *A*. NRM. *B*. After demagnetization. Closed and open circles represent sites with, respectively, positive and negative inclinations.

The mean magnetic vectors for 25 statistically significant sites are plotted on an equal-area (Schmidt) stereographic projection for both untreated sites and demagnetized sites (fig. 4). The untreated NRM values display substantial dispersion, and a majority of the sites have positive inclinations. The stereographic plot of the same sites after optimal demagnetization displays two clusters of sites, which are antipodal and consist of either northerly declinations and positive inclinations (interpreted to be of normal polarity) or southerly declinations and negative inclinations (interpreted to be of reversed polarity).

The point dispersion and the inconsistencies of declinations and inclinations that are seen for the untreated sites are characteristic for sites with overprinted, secondary components of the NRM. The clustering of sites after demagnetization is the result of partial to complete removal of this component and the remaining NRM suggest isolation of a characteristic magnetization that probably resulted from either DRM or CRM.

The mean values for the three normal sites are declination 358.6°, inclination 44.63° (table 1). The mean values for the 22 reversed sites are declination 177.57°, inclination

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TABLE 1

PALEOMAGNETIC DATA FOR STATISTICALLY SIGNIFICANT (CLASS I) SITES OF UNAMBIGUOUS POLARITY FOR THE BLACK PEAKS FORMATION

Site No.	Optimum Demag.	I	D	N	R	k	α_{95}	Polarity
1	60 mT	−25.01	185.76	3	2.97	89.50	13.11	R
2	400°C	−28.83	168.44	3	2.97	64.11	15.52	R
4	40 mT	−28.19	180.00	3	2.95	43.50	18.92	R
7	50 mT	−37.70	181.76	3	2.82	11.40	38.34	R
8	25 mT	−18.06	183.23	3	2.98	106.06	12.03	R
10	500°C	51.00	2.50	3	2.97	57.21	16.45	N
12	25 mT	−13.42	174.18	3	2.91	21.63	27.18	R
14	45 mT	−27.98	169.47	4	3.91	34.74	15.81	R
18	400°C	48.83	3.46	4	3.74	11.43	28.41	N
20	45 mT	−16.12	172.15	3	2.95	36.93	20.58	R
22	45 mT	−16.48	167.60	3	2.96	56.83	16.51	R
23	60 mT	−8.69	191.43	3	2.98	85.25	13.44	R
25	45 mT	−31.23	202.67	3	2.98	108.53	11.89	R
28	25 mT	−17.84	172.60	3	2.95	27.92	23.78	R
30	45 mT	−41.15	187.42	3	2.91	23.49	26.02	R
31	25 mT	−7.17	183.05	3	2.91	22.83	26.42	R
32	25 mT	−16.62	175.47	3	2.98	175.47	12.71	R
52	400°C	−27.85	188.73	4	3.64	8.52	33.41	R
57	25 mT	33.63	351.79	3	2.98	102.73	12.23	N
62	60 mT	−17.10	180.54	3	2.94	38.53	20.14	R
63	25 mT	−29.41	169.78	3	2.96	47.21	18.15	R
64	60 mT	−39.35	152.08	3	2.98	91.91	12.93	R
65	60 mT	−32.00	154.93	3	2.85	13.69	34.69	R
67	45 mT	−34.94	203.53	3	2.81	10.63	39.85	R
69	25 mT	−41.08	158.41	3	2.80	10.24	40.69	R
All	Reversed	−25.82	177.57	22	21.23	27.26	6.04	R
All	Normal	44.63	358.60	3	2.97	58.59	16.25	N
All Normal and Reversed		−28.12	357.56	25	24.04	25.02	5.90	...

−25.82° (table 1). As depicted in figure 5, the two mean values for the normal and reversed (inverted) sites represent the same population at the 95% confidence level. This antipodal distribution indicates that any secondary magnetization preferentially affecting either the normal or reversed sites has been successfully removed by demagnetization. The formational mean value for the 25 statistically significant sites (with reversed inverted) is declination 357.56, inclination 28.12 (table 1).

Barnes and Butler (1980) discussed the importance of the early Tertiary apparent polar wander path for an understanding of the tectonic evolution of North America. Because of the paucity of data points for this time interval, a paleomagnetic pole position was calculated for all Class I sites from the Black Peaks Formation, which yielded a value of lat. 75.48°N, long. 86.24°E (N = 25, dp = 3.54, dm = 6.46). The pole latitude is consistent with previously published Paleocene values for North America (e.g., Barnes and Butler 1980; Jacobson et al. 1980). However, the pole longitude for the Black Peaks Formation is significantly divergent, i.e., it lies far eastward, from previous data points for the North American Paleocene (refs. cited

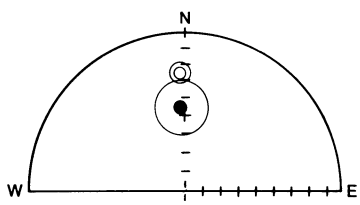


FIG. 5.—Stereographic projection of mean of all normal and reversed (inverted) sites represented by, respectively, closed and open circles with corresponding α_{95} circles of confidence. Cross represents inclination of the axial geocentric dipole field expected for the present-day latitude of the Black Peaks Formation.

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above). The probable explanation for this discordance of longitude is the seemingly low inclinations obtained for many of the sites of reversed polarity (Channell, pers. comm. 1982; also see fig. 5 and discussion above). Therefore, the Black Peaks formational pole position should not be added to the Paleocene data set for North America nor should it be used for any broad-scale tectonic interpretations.

MAGNETIC POLARITY ZONATION

The hierarchical classification of Opdyke et al. (1977) was used to determine the magnetic polarity zonation of the Black Peaks sections. Class I sites are those of unambiguous polarity that consist of three or more samples and are statistically significant at the 95% confidence level. The 25 sites discussed above fall into Class I. Class II sites consist of two samples per site but each displays consistent vectors that indicate either normal or reversed polarities. The 11 Class II sites mostly consist of poorly indurated samples from which only two specimens remain after transport from the field and preparation in the laboratory. In addition, two sites were determined to be Class III, i.e., consisting of three samples of which two have concordant directions. Therefore, of the 53 sites originally sampled, a total of 38 were ultimately used to determine the magnetic polarity zonation of the Black Peaks Formation. These include 29 sites along the type section and nine sites along the reference section 1 (see below).

The 182 m magnetic polarity zonation along the type section extends from a few meters below the base of the Black Peaks Formation to about 10 m above the base of the overlying Hannold Hill Formation (fig. 6). The magnetic polarity zonation of the type section consists of four reversed magnetozones (R1 to R4) and four normal magnetozones (N1 to N4). The longest magnetic polarity zone, R1, extends from 0 m to 60 m, is based on 12 sites, and is overlain by a relatively short normal magnetozone (N1). Zone N1, which is based on one site, is about 15 m thick (transitions are placed midway between sites of opposite polarity). In order to demonstrate that N1 does not represent a low blocking temperature secondary overprint, an extensive thermal demagnetization regime was done for each of the three specimens from site 10 (fig. 6). In each case, the polarity remained normal to over 600°C (also see fig. 3, table 1). These data strongly suggest that the characteristic component of magnetization had been successfully isolated. Zone N1 is overlain by an intermediate length zone R2, which is based on three sites and extends from about 75 m to 95 m. Zone R2 is overlain by a short normal zone N2, based on one site, that extends from about 95 m to 100 m. As also discussed above (see e.g., fig. 3), the magnetic characteristics of this site indicate normal polarity even during thermal demagnetization up to 640°C. A short reversed magnetozone R3, which extends from 100 m to 105 m, is based on one site. Zone N3, which is based on two sites, extends from about 105 m to 115 m. The long zone R4, based on eight sites, extends from 115 m to 165 m. The top of the measured zone N4, based on one site, extends to the top of the measured section.

The magnetic polarity column for the reference section 1 located on eastern Tornillo Flat (fig. 7) contains approximately 65 m of the lower Black Peaks Formation. As will be discussed below, this section is important because it contains a relatively rich fossil locality. Reference section 1 extends about 15 m below the Black Peaks into the underlying Javelina Formation. All nine sites of reference section 1 are of reversed polarity (fig. 7). This magnetostratigraphy consists of two reversed magnetozones (R-1 and R-2) that are separated by an indeterminant zone (I-1). It was not possible to sample in the 30 m I-1 zone because this part of the section consisted of very poorly indurated sandstone or was completely covered by alluvium.

MAMMALIAN BIOSTRATIGRAPHY

The mammalian faunas of the Black Peaks Formation have been extensively discussed by Wilson (1967) and Schiebout (1974, 1980). Schiebout describes several mammalian faunal levels keyed to the Black Peaks type section and one eastern Tornillo Flat locality found along reference section 1. The biochronology of these faunal levels includes the late Torrejonian (middle Paleocene), Tiffanian (late Paleocene), and Clarkforkian (earliest Eocene, *sensu* Rose 1981). In addition, a Wasatchian faunule ("TT Jacks" locality) is known from the lower part of the Hannold Hill Formation on western Tornillo Flat (Wil-

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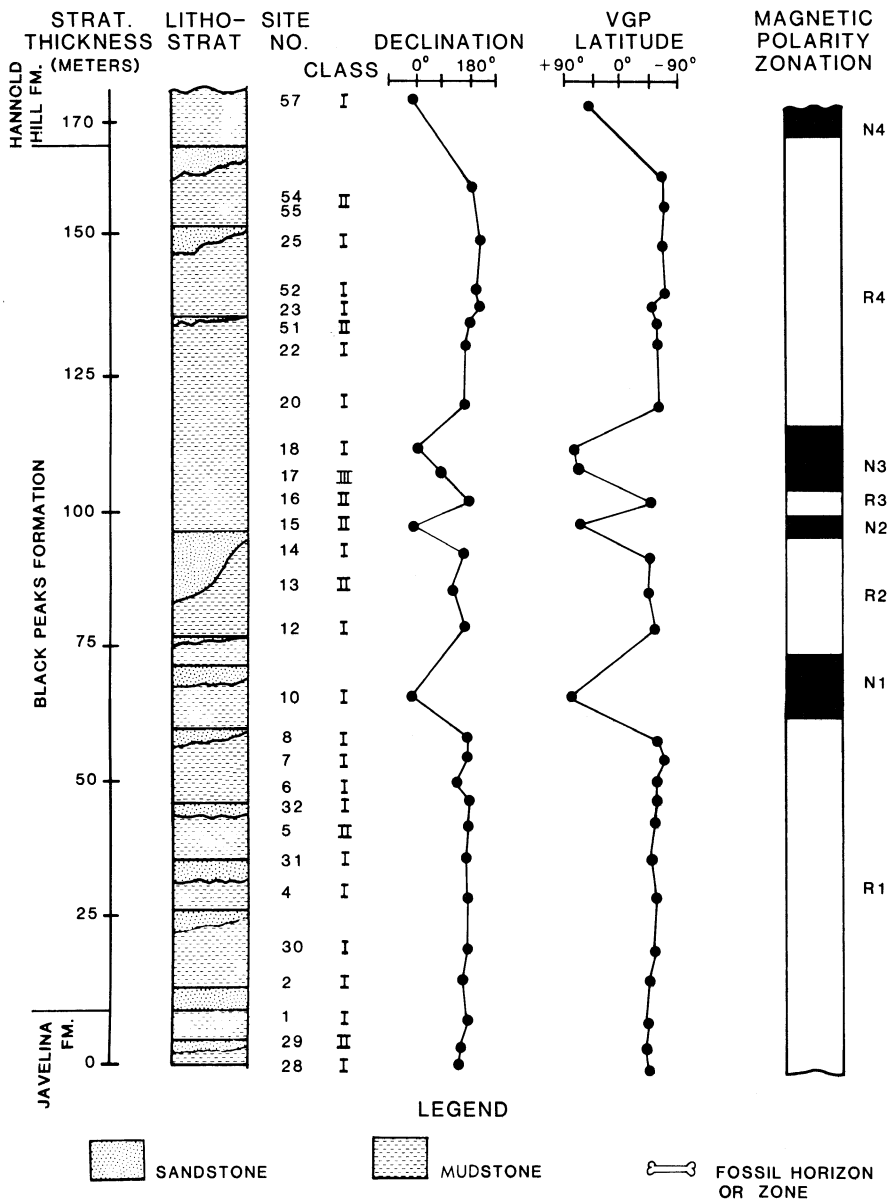


FIG. 6.—Stratigraphic and paleomagnetic data for the type section of the Black Peaks Formation, western Tornillo Flat.

son, *in* Maxwell et al. 1967; Hartnell 1978, 1980). This occurrence is important because it is in direct superposition with the faunal levels from the type Black Peaks Formation (also see Discussion below). The stratigraphic distribution of these mammalian faunas is presented in table 2.

The lowest fossil level is located 23 m above the base of the type section (fig. 8). This locality contains two species, *Promioclaenus acolytus* and *Caenolambda,* that are not found higher in the formation. *P. acolytus* is only known from the Torrejonian. The Big Bend *Caenolambda* most closely resembles *C. jepseni,* which biochronologically approximates the Torrejonian-Tiffanian boundary (Schiebout 1974). Therefore, the fossils from the 23 m level are shown in figure 8 as Torrejonian-Tiffanian.

The most prolific eastern Tornillo Flat fos-

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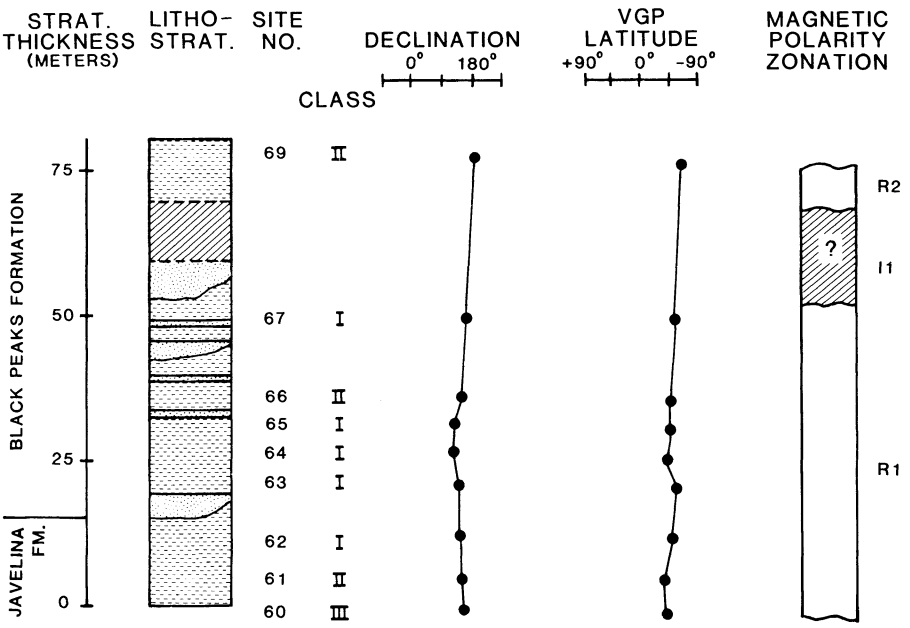


FIG. 7.—Stratigraphic and paleomagnetic data for reference section 1 of the Black Peaks Formation, eastern Tornillo Flat. See figure 6 for symbols.

sil locality T-2 (see Wilson, *in* Maxwell et al. 1967) is located about 43 m above the Javelina-Black Peaks contact. This fossil level was originally thought to be Torrejonian in age, but is now considered to be slightly younger, i.e., Tiffanian (Schiebout 1974). Several genera of multituberculates and condylarths are found at this locality (table 2). At the 43 m level in the type section of western Tornillo Flat, a pantodont, *Titanoides zeuxis,* is considered Tiffanian in age.

In the type section, two relatively fossiliferous localities, which are referred to as Ray's Bonebed and Ray's Annex (Schiebout 1974), are located at the 49–59 m level. The faunas of Ray's Annex and Ray's Bonebed include two genera of primates, one genus of taeniodont, one genus of arctocyonid, and several genera of condylarths and pantodonts (table 2). Ray's Annex and Ray's Bonebed are considered to be middle Tiffanian (Schiebout 1974).

The most fossiliferous locality in the Black Peaks Formation is Joe's Bonebed, which lies 79–85 m above the base of the type section. Several genera of primates, multituberculates, insectivores, arctocyonid, and condylarths are found at this level (table 2). This fossil assemblage is considered to be Tiffanian in age (Schiebout 1974). Faunal remains of a taeniodont, *Psittacotherium* or *Lampadophorus,* and an undescribed genus of arctocyonid were found 20 m above Joe's Bonebed at the 105 m level. These remains are also Tiffanian in age (Schiebout 1974).

Three teeth of *Hyracotherium angustidens* and a single molar of the pantodont *Barylambda* were discovered at the 137 m level (Schiebout 1974). *Hyracotherium* suggests a Wasatchian age (*sensu* Rose 1981) and *Barylambda* is late Tiffanian and Clarkforkian in age. The only other co-occurence of these two taxa is from Baja California (Morris 1968), which was considered Tiffanian, Clarkforkian, or Wasatchian with a preference for the Clarkforkian. A reasonable assignment to a land mammal age for these specimens is Clarkforkian (Schiebout 1974).

Gingerich (1976, fig. 15) made an approximate correlation of the Black Peaks plesiadapids with those from more northern localities, e.g., the Bighorn Basin sequence, Wyoming. This correlation places Ray's Bonebed at a biostratigraphic level equivalent to the *Plesiadapis rex* biochron and Joe's Bonebed at the *Plesiadapis simonsi* biochron. Based on this correlation and its relationship to Gingerich's land mammal bio-

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TABLE 2

STRATIGRAPHIC DISTRIBUTION OF EARLY TERTIARY (MEDIAL PALEOCENE THROUGH EARLY EOCENE) MAMMALS COLLECTED FROM THE BLACK PEAKS FORMATION AND OVERLYING HANNOLD HILL FORMATION ON WESTERN TORNILLO FLAT AND BLACK PEAKS FORMATION ON EASTERN TORNILLO FLAT

	Level above base of Black Peaks Formation (in meters):							
Taxon	23 m	43[a] m	43 m	49–59 m	79–85 m	105 m	137 m	189–[b] 215 m
MULTITUBERCULATA								
Ptilodus mediaevus	x	x			x			
Ectypodus musculus		x			x			
Parectypodus sloani		x			x			
Mimetodon silberlingi		x			x			
Neoplagiaulax douglassi		x						
Mesodma sp.		x						
INSECTIVORA								
? *Palaeictops sp.*					x			
Jepsenella sp.					x			
PRIMATES								
? *Zanycteris sp.*					x			
Pleisiadapis gidleyi				x	x			
Chiromyoides caesor					x			
Phenacolemur frugivorus					x			
Navajovius kohlhaasae		x		x	x			
Phenacolemur cf. *P. praecox*								x
TAENIODONTA								
Psittacotherium or *Lampadophorus*		x		x	x	x		
CARNIVORA								
Tricentes truncatus				x	x			
Arctocyon ct. *A. ferox*					x			
Undescribed genus and species of Arctocyonid						x		
CONDYLARTHRA								
Promioclaenus acolytus	x							
Haplaletes disceptatrix					x			
Protoselene opisthacus				x				
Phenacodus grangeri	x			x	x			
Phenacodus cf. *P. matthewi*	x	x		x	x			
Ectocion cf. *E. montanensis*				x	x			
Periptychus superstes	x	x		x				
Phenacodus primaevus								x
Hyopsodus meticulus								x
Hyopsodus cf. *H. wortmani*								x
PANTODONTA								
Caenolambda sp.	x							
Barylambda jackwilsoni				x				
Barylambda sp.							x	
Titanoides zeuxis			x	x				
Coryphodon sp.								x
PERISSODACTYLA								
Hyracotherium angustidens							x	
Hyracotherium vasacciense								x
RODENTIA								
Paramys excavatus								x

SOURCES.—Adapted from Schiebout 1974; Hartnell 1980, 1981.

[a] On eastern Tornillo Flat.

[b] TT Jack's locality, Hannold Hill Formation, western Tornillo Flat. Not shown in figures 8 and 9.

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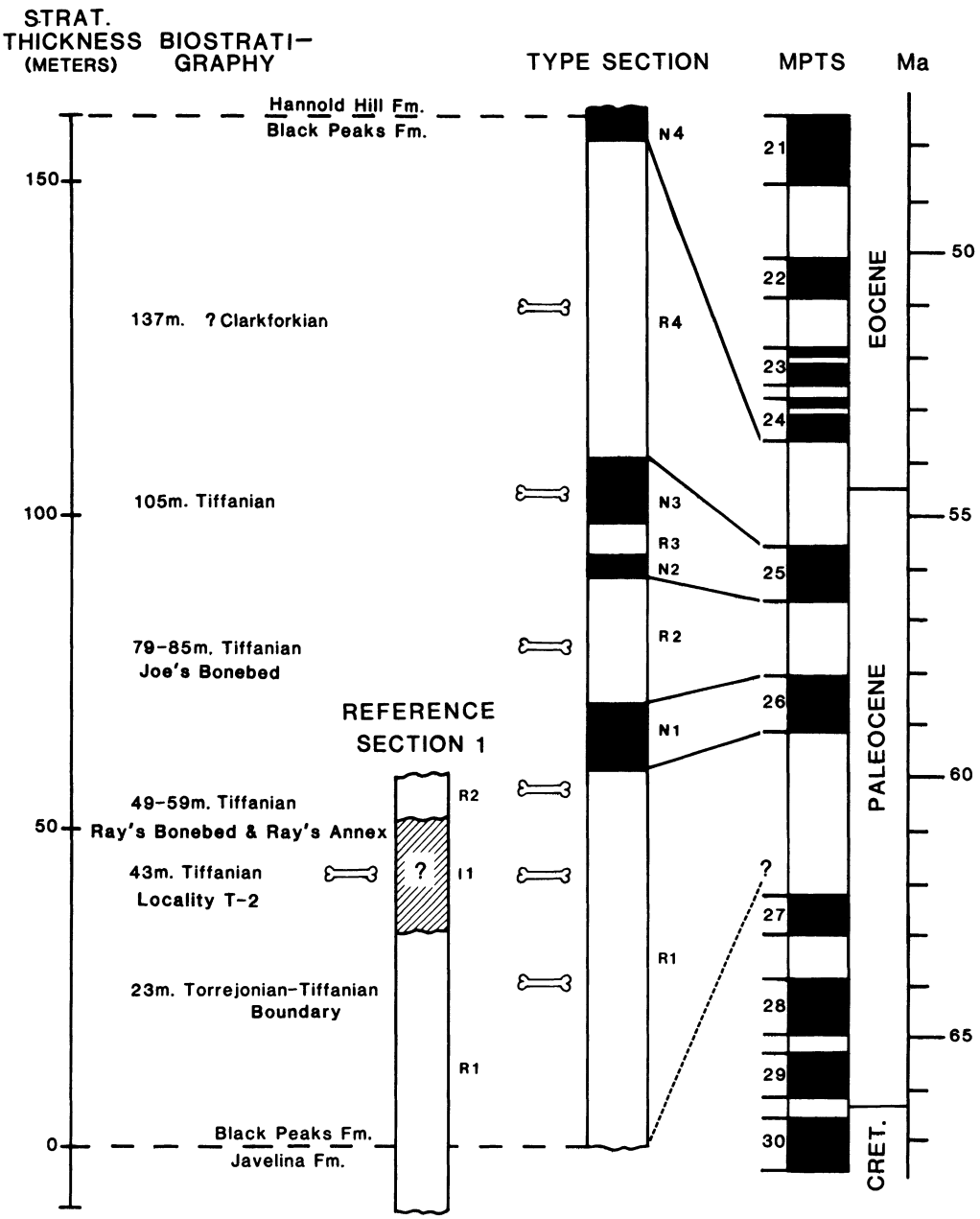


FIG. 8.—Biostratigraphy, biochronology, and correlation to the MPTS of the type section and reference section 1 of the Black Peaks Formation. See figure 6 for symbols.

chrons, Ray's Bonebed is equivalent to the Ti_3 biochron, while Joe's Bonebed is equivalent to the Ti_5 biochron (fig. 9). Based on correlation to the absolute time scale, Gingerich (1980) estimated that the Ti_3 and Ti_5 biochrons can be dated, respectively, at about 58 and 56 Ma.

MAGNETIC POLARITY CHRONOLOGY

The correlation of the magnetic polarity stratigraphy of the Black Peaks Formation to the standard Magnetic Polarity Time Scale (MPTS) is based on the approximate biostratigraphic age of the Black Peaks faunal succession and the correlation of the Black

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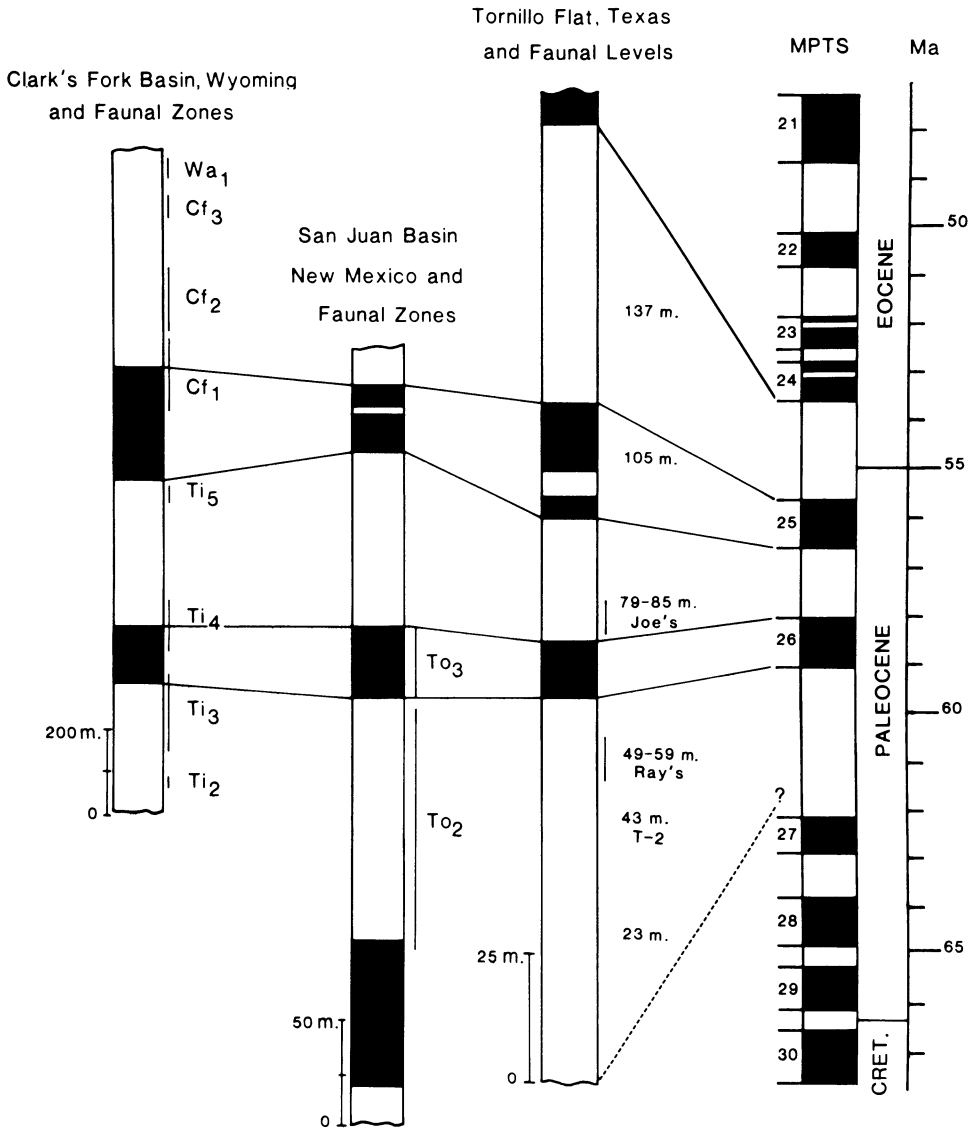


FIG. 9.—Correlation of the Big Bend early Tertiary sequences on Tornillo Flat with other roughly contemporaneous sequence with associated chronological control, i.e., Clark's Fork Basin, Wyoming and the San Juan Basin, New Mexico (see text for references). The abbreviations for mammalian faunal zones (see Gingerich 1980) are as follows: To, Torrejonian; Ti, Tiffanian; Cf, Clarkforkian; Wa, Wasatchian. For the Tornillo Flat section, the important faunal zones are indicated as follows: T-2, Ray's (for Ray's Bonebed and Ray's Annex), and Joe's (for Joe's Bonebed).

Peaks magnetic reversal pattern to that of the MPTS (following Lowrie and Alvarez 1981). The biochronology of the Black Peaks faunas suggests that the formation has an age range of latest middle Paleocene (Torrejonian-Tiffanian boundary) to earliest Eocene (Clarkforkian). The transition between Torrejonian-Tiffanian faunas occurs at approximately 60 Ma (Gingerich 1980), Tiffanian faunas range in age from about 60 Ma to about 56 Ma, and the transition between Tiffanian-Clarkforkian faunas occurs at about 56 Ma (fig. 9). Based on these biostratigraphic age constraints, the faunal succession in the Black Peaks Formation, and the MPTS of Lowrie and Alvarez (1981), the Black Peaks magnetozones should be no older than chron 27 and no younger than chron 24.

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The 23 m fossil locality level, where the faunas approximate the Torrejonian-Tiffanian boundary, is located in the lower third of the lowest reversed polarity magnetozone, R1 (figs. 8, 9). The eastern Tornillo locality occurs in a zone of indeterminate polarity bracketed by zones of reversed polarity (fig. 8). In the type section the *Titanoides* locality (early Tiffanian) occurs in the upper third of R1. The lowest normal polarity magnetozone, N1, is bracketed at 49–59 m by Ray's Bonebed and Joe's Bonebed at 79–85 m (middle-late Tiffanian). The 105 m faunal site (late Tiffanian) is located in the middle of the N3 magnetozone. The *Hyracotherium* and *Barylambda* locality (probably Clarkforkian) is located approximately in the center of the R4 magnetozone at 137 m.

This biochronology and its relationship to the Black Peaks magnetic polarity stratigraphy, when compared to the MPTS, allows correlation of the Black Peaks magnetozones. The age constraints, as imposed by the Black Peaks fauna, imply that R1 is closest in age to the relatively long reversal below chron 26 (figs. 8, 9), N1 correlates to chron 26, and R4 is closest in age to the reversal below chron 24 (fig. 9). Therefore, R2 is correlated to the reversal below chron 25, N2, R3, and N3 are correlated to chron 25, and N4 is correlated to chron 24. It should be noted here that R3 is interpreted to represent a short reversed event not recorded in the MPTS. This overall correlation is further supported by the fact that the sequential pattern of normal and reversed polarity magnetozones is correlative to the pattern (i.e., the relative lengths) of these anomalies on the MPTS (figs. 8, 9).

In summary, based on the magnetozone pattern, the Black Peaks Formation is best correlated to the reversals above chron 27 up to the base of chron 24 (figs. 8, 9), or an absolute interval of time of about 62 Ma to 53 Ma. Any other correlation of magnetic reversal patterns would require a great amount of fluctuation in the rate of deposition. However, Schiebout (1970, 1974, 1980, 1981, pers. observ.) and Rapp (1982) support the conclusion that accumulation rates of the Black Peaks strata were relatively unchanged throughout the depositional history of this formation.

In both Texas and Wyoming the general polarity patterns are similar. Also the correspondence of the land mammal ages indicate Tiffanian faunas occur in the reversed zone below chron 26. The Clarkforkian faunal levels in both areas occur in chron 25 and up into the reversed zone below chron 24. In the Bighorn Basin, the presence of Wasatchian mammals at the top of the reversed zone below chron 24 is not inconsistent with the presence of a faunule of similar age in the Hannold Hill Formation (not illustrated in figs. 8 and 9, but this locality, a small outcrop surrounded by Recent alluvium, is estimated at 189–211 m above the base of the Black Peaks Formation, see table 2).

On the other hand, there is a discrepancy between the correlation of the Paleocene Black Peaks Formation and the Nacimiento Formation of the San Juan Basin, New Mexico (Butler et al. 1977; Lindsay et al. 1980). The general pattern of magnetozones is similar in the two regions, i.e., several relatively long reversed zones punctuated by shorter zones of normal polarity. Considering only the time interval of mutual overlap, the lower normal magnetozone, which contains Torrejonian mammals in the New Mexico sequence, is correlated to chron 26 of the MPTS. If this correlation is correct, then a problem arises in the putative homotaxis of this land mammal age, because in the Wyoming and Texas sequences, chron 26 is characterized by Tiffanian mammals. Possible explanations for this seeming discrepancy will be discussed below.

IMPLICATIONS OF MAGNETIC POLARITY CHRONOLOGY

Sedimentation Rates.—The correlation of the relative lengths of the Black Peaks magnetozones to the MPTS suggests that the overall rate of accumulation by vertical aggradation was slow and fairly constant over the span of time represented by the Black Peaks type section. The average rate of accumulation for a stratigraphic sequence can be calculated by dividing the lithostratigraphic thickness, within one or more magnetozones, as indicated by the MPTS. The age dates and durations are derived from Lowrie and Alvarez (1981).

The average rates of accumulation by vertical aggradation for the major Black Peaks magnetozones that have both upper and lower contacts are presented in table 3. The

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TABLE 3

CALCULATION OF SEDIMENTATION RATES FOR THE TYPE SECTION OF THE BLACK PEAKS FORMATION

Magnetozones	Chron	Duration My	Stratigraphic Thickness	Rate of Accumulation cm/1000y
N1	26	.8	12.0	1.5
R2		1.49	18.0	1.21
N2, R3, N3	25	.90	20.0	2.22
R4		2.15	48.0	2.23
Total or Mean Values		5.34	98.0	1.83

overall average vertical aggradation between the base of magnetozone N3 (chron 24) and the base of magnetozone N1 (chron 26), which is the total of all complete Black Peaks magnetozones, is 1.83 cm/1000 years. The mean values presented in table 3 indicate that the rate of accumulation appears to have been slightly slower during chron 26 and up to chron 25 time, and that accumulation was slightly more rapid during chron 25 and up to chron 24 time. The rate of accumulation during the reversal prior to chron 26 time (R1) cannot be calculated, because this magnetozone has no definitive base, but appears to be slightly greater relative to the rate during chron 26 time because this magnetozone appears to be longer than the other polarity zones when compared to the MPTS (fig. 9). Thus, vertical aggradation of the Black Peaks Formation perhaps was slightly more rapid for the lower portion of the formation (late Torrejonian-early Tiffanian), slowed through the middle portion of the formation (mid-late Tiffanian), and became more rapid again through the upper portion of the formation (late Tiffanian-Clarkforkian). If indeed these variations in accumulation rates are significant, they probably resulted from slight changes in subsidence rates and/or base levels. For the entire Black Peaks Formation, average vertical aggradation probably averaged about 2 cm/1000 years. Despite this variation, the rates of sedimentation calculated for different parts of the Black Peaks are relatively constant, i.e., they are within the same order of magnitude (ca. 1 cm/1000 yrs.).

Local Chronology.—The magnetic polarity stratigraphy of the Black Peaks Formation demonstrates that the Javelina-Black Peaks and Black Peaks-Hannold Hill formational contacts do not approximate the Cretaceous-Paleocene and Paleocene-Eocene time boundaries (see Maxwell et al. 1967). The base of the Black Peaks Formation lies within magnetozone R1, which is equivalent to the reversal between chrons 26 and 27. Based on this correlation, the base of the Black Peaks can be no older than middle Paleocene (Torrejonian). Maxwell et al. (1967) and Lawson (1972) report the presence of dinosaurs from the Javelina Formation. In addition, recent field work has recovered mammals from near the top of the Javelina Formation (Stanhardt, pers. comm.). Therefore, as also predicted from the present study, the Cretaceous-Tertiary boundary should occur in the upper part of the Javelina Formation which directly underlies, and is stratigraphically continuous with, the type section of the Black Peaks Formation. The unconformity at the base of a fluvial sandstone, reported by Maxwell et al. (1967) as the lower boundary of the Black Peaks Formation, is an erosion surface not greater than beneath the other fluvial sandstones within the formation. With Maastrichtian dinosaurs reported from other parts of the Javelina Formation (Lawson 1972), the Javelina sequence should overlap with those presented in other studies such as the San Juan Basin of New Mexico (Butler et al. 1977; Lindsay et al. 1978), the "Dragonian" (actually early Torrejonian) of Utah (see Tomida and Butler 1980; Tomida 1981),

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and the late Cretaceous (Maastrichtian) Hell Creek Formation and early Paleocene (Puercan) Tullock Formation of Montana (Archibald et al. 1982).

The Black Peaks-Hannold Hill contact is located at the level of magnetozone N4, which is equivalent to the base of chron 24. The Paleocene-Eocene epoch boundary at about 55 Ma is located between chrons 24 and 25 on the MPTS. Thus the Paleocene-Eocene boundary is not located at the Black Peaks-Hannold Hill contact but approximately the 125 m level of the type section in zone R4 (fig. 9). This conclusion is also supported by the presence of Clarkforkian mammals in R4 that, based on revised interpretations, are early Eocene in age (Rose 1981). Thus, based on correlation to the absolute time scale, the Black Peaks Formation has an age range that is no older than about 62 Ma and no younger than about 53 Ma, resulting in a duration of no more than about 9 m.y. The formation is younger than originally suggested by Maxwell et al. (1967), and thus the top of the Javelina Formation and the base of the Hannold Hill Formation are younger than originally thought and do not approximate the Cretaceous-Paleocene and Paleocene-Eocene contacts.

With regard to the faunal succession, based on the Black Peaks magnetic polarity chronology, the fossil levels described above have the following approximate ages: the 23 m fossil level has an age of 61.0 Ma; the 43 m localities have an age of 59.5–60.0 Ma; Ray's Annex and Ray's Bonebed have an age of 59.0–59.5 Ma; Joe's Bonebed has an age of 57.8–58.2 Ma; the 105 m fossil level has an age of 55.8–56.2 Ma; and the 137 m locality has an age of 54.9–55.3 Ma. These age dates also are consistent with the calibration of the respective land mammal ages for these levels.

Intracontinental Correlations.—The Black Peaks magnetic polarity stratigraphy allows for correlation and comparison to other mammal-bearing sedimentary sequences. As previously discussed, the Black Peaks magnetic polarity reversal sequence and faunal succession correlate well to that of the Clark's Fork Basin (fig. 9). By comparing the Black Peaks and Clark's Fork biostratigraphy, relative to their respective magnetic polarity stratigraphies, Ray's Bonebed and Ray's Annex are equivalent to the Clark's Fork Basin Ti_3 biochron; Joe's Bonebed is equivalent to the Ti_4 biochron; the 105 m level is equivalent to the Cf_1 biochron. Comparisons of the 23 m and 43 m fossil levels to the Clark's Fork Basin's biochrons are difficult to make because of the indeterminant base of the lowermost magnetozone of both areas.

The biostratigraphy of the Black Peaks and Clark's Fork Basin does not correspond to the biostratigraphy of the San Juan Basin as it is correlated to the MPTS (Butler et al. 1977; Lindsay et al. 1978; Lindsay et al. 1981). Since the publication of Butler et al. (1977), there have been a series of publications that support or refute the original correlation of magnetozonation in the San Juan Basin (see references in Lucas and Schoch 1982; Lindsay et al. 1982; and earlier references cited therein). At first glance one might appeal to the idea of a major unconformity in the Cretaceous part of the San Juan sequence (e.g., Alvarez and Vann 1979; Lucas and Rigby 1979; Lucas and Schoch 1982). However, this does not resolve the problem because if there were a hiatus, then magnetozones and Torrejonian faunal levels would be shifted even higher, which is still in contrast to the situation in Texas and New Mexico. Another possibility is that chron 25 is as identified in the San Juan Basin is actually chron 26 as correlated to the MPTS. With regard to paleontological evidence, the only way one could support the hypothesis of apparent "faunal shifts" due to actual major unconformities would be to invoke a hiatus in the reversed zone between chrons 25 and 26 as correlated in the San Juan Basin. However, there is no lithostratigraphic evidence to support this assertion (although see Fassett 1979). Another hypothesis could be biological, i.e., some ecological or geographic differences in faunas. If this were the case, then Torrejonian mammalian faunas of the San Juan Basin were contemporaneous with Tiffanian mammalian faunas of the Clark's Fork and Black Peaks areas. This distribution is not likely to be the result of immigration of Torrejonian faunas occurring later in the San Juan Basin than in the other areas because the San Juan Basin is geographically located between the Black Peaks area and the Clark's Fork Basin area. Thus an unambiguous correlation of biostratigraphy and magnetic polarity stratigraphy cannot presently be made for these three areas. This problem will have to be resolved with further studies

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of these and other early Tertiary land mammal fossil localities.

In summary, this paper presents an interdisciplinary chronology for the early Tertiary sediments and mammals from the Black Peaks Formation. This research represents the first phase of a detailed and comprehensive study of the combined late Cretaceous to early Tertiary vertebrate-bearing sequence in Big Bend National Park. Future studies will be directed toward additional refinements in the geochronology of this sequence as well as comparisons with similar deposits elsewhere in North America.

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